No.	Status	Date
LNDE-RPT-20220906-033-DS-6-SG-102	Issued	2022-09-16

UT Thickness Data Sheet - 6-SG-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 6-SG-102, located within the ATU-100. The scope of work involves the recording of wall thickness at designated points to monitor asset integrity and corrosion rates.

Marcus Chen, acting as the lead UT technician and SNT-TC-1A UT Level II for Larkhall NDE Services Inc., performed the measurements and is responsible for the accuracy of the recorded data.

Technical Specifications

The subject line, 6-SG-102, is a sour gas inlet constructed from A106 Gr.B with an outer diameter of 168.3 mm and a nominal thickness of 7.11 mm. The design limits for this line are 1965 kPa(g) and 66 °C.

The minimum allowable thickness (t-min) is 3.56 mm. Any reading recorded below this retirement thickness is subject to a non-conformance report.

Execution and Methodology

Thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 2. This procedure includes specific mandates for insulation removal, reinstatement, and inspection-port requirements as part of the CUI program.

The inspection was conducted in accordance with the standard ASTM E797. The measurements were executed to support the 3-year RBI survey for Class 1 sour gas, regenerator overhead and rich amine.

Reference and Reporting

The data contained herein is cited in LNDE-RPT-20220906-033-R0, the comprehensive report for the UT Thickness Survey of the ATU-100 sour gas, regenerator overhead, and rich amine systems.

This data sheet serves as the primary record for the thickness readings of 6-SG-102, facilitating the calculation of short-term and long-term corrosion rates and the determination of remaining life.

Thickness readings, measured 2022-09-16

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2019-10-18	5.93	5.82	0.049	0.053	42.6	2025-09-16	
6-SG-101-02	7.11	3.56	2019-10-18	5.51	5.42	0.061	0.052	30.3	2025-09-16	
6-SG-101-03	7.11	3.56	2019-10-18	5.56	5.29	0.067	0.088	19.7	2025-09-16	
6-SG-101-04	7.11	3.56	2019-10-18	5.34	5.19	0.072	0.059	22.8	2025-09-16	
6-SG-101-05	7.11	3.56	2019-10-18	5.04	4.72	0.091	0.100	11.6	2025-09-16	
6-SG-101-06	7.11	3.56	2019-10-18	5.56	5.55	0.057	0.050	34.7	2025-09-16	
6-SG-101-07	7.11	3.56	2019-10-18	6.24	6.16	0.027	0.027	>50	2025-09-16	
6-SG-101-08	7.11	3.56	2019-10-18	5.77	5.57	0.066	0.065	30.6	2025-09-16	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2019-10-18	5.05	4.77	0.089	0.099	12.3	2025-09-16	
6-SG-102-02	7.11	3.56	2019-10-18	5.10	4.64	0.093	0.120	9.0	2025-09-16	
6-SG-102-03	7.11	3.56	2019-10-18	4.92	4.69	0.089	0.080	12.8	2025-09-16	
6-SG-102-04	7.11	3.56	2019-10-18	4.90	4.62	0.089	0.085	11.9	2025-09-16	
6-SG-102-05	7.11	3.56	2019-10-18	6.07	5.96	0.044	0.035	>50	2025-09-16	
6-SG-102-06	7.11	3.56	2019-10-18	6.03	5.79	0.054	0.041	41.6	2025-09-16	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2019-10-18	7.29	6.83	0.064	0.152	19.9	2025-09-16	
4-RA-101-02	8.56	3.81	2019-10-18	7.39	6.79	0.066	0.174	17.1	2025-09-16	
4-RA-101-03	8.56	3.81	2019-10-18	6.64	5.98	0.099	0.253	8.6	2025-09-16	
4-RA-101-04	8.56	3.81	2019-10-18	7.54	7.21	0.054	0.120	28.4	2025-09-16	
4-RA-101-05	8.56	3.81	2019-10-18	7.18	6.73	0.071	0.165	17.7	2025-09-16	
4-RA-101-06	8.56	3.81	2019-10-18	6.94	—	—	—	—	—	inaccessible – no reading
4-RA-101-07	8.56	3.81	2019-10-18	7.68	7.50	0.041	0.097	38.0	2025-09-16	
4-RA-101-08	8.56	3.81	2019-10-18	6.45	5.55	0.118	0.267	6.5	2025-09-16	
4-RA-101-09	8.56	3.81	2019-10-18	7.68	7.37	0.045	0.115	30.9	2025-09-16	
4-RA-101-10	8.56	3.81	2019-10-18	7.55	7.19	0.056	0.126	26.8	2025-09-16	
4-RA-101-11	8.56	3.81	2019-10-18	6.66	6.07	0.098	0.232	9.7	2025-09-16	
4-RA-101-12	8.56	3.81	2019-10-18	7.03	6.41	0.213	0.213	12.2	2025-09-16	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2019-10-18	7.58	7.42	0.042	0.091	39.7	2025-09-16	
4-RA-102-02	8.56	3.81	2019-10-18	6.96	6.43	0.081	0.184	14.3	2025-09-16	
4-RA-102-03	8.56	3.81	2019-10-18	7.05	6.56	0.075	0.150	18.3	2025-09-16	
4-RA-102-04	8.56	3.81	2019-10-18	7.04	6.64	0.072	0.164	17.3	2025-09-16	
4-RA-102-05	8.56	3.81	2019-10-18	7.06	6.56	0.075	0.168	16.3	2025-09-16	
4-RA-102-06	8.56	3.81	2019-10-18	7.15	6.65	0.070	0.164	17.3	2025-09-16	
4-RA-102-07	8.56	3.81	2019-10-18	7.30	6.85	0.065	0.141	21.6	2025-09-16	
4-RA-102-08	8.56	3.81	2019-10-18	6.40	5.86	0.099	0.226	9.1	2025-09-16	
4-RA-102-09	8.56	3.81	2019-10-18	6.78	6.42	0.080	0.170	15.4	2025-09-16	
4-RA-102-10	8.56	3.81	2019-10-18	7.36	6.86	0.172	0.172	17.8	2025-09-16	
4-RA-102-11	8.56	3.81	2019-10-18	7.72	7.33	0.074	0.096	36.8	2025-09-16	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2019-10-18	9.83	9.55	0.055	0.112	44.4	2025-09-16	
6-RA-103-02	10.97	4.57	2019-10-18	8.61	7.97	0.118	0.261	13.0	2025-09-16	
6-RA-103-03	10.97	4.57	2019-10-18	9.49	8.95	0.077	0.168	26.0	2025-09-16	
6-RA-103-04	10.97	4.57	2019-10-18	8.22	7.30	0.139	0.305	9.0	2025-09-16	
6-RA-103-05	10.97	4.57	2019-10-18	8.33	7.43	0.133	0.311	9.2	2025-09-16	
6-RA-103-06	10.97	4.57	2019-10-18	9.79	9.45	0.059	0.132	37.0	2025-09-16	
6-RA-103-07	10.97	4.57	2019-10-18	9.59	9.15	0.069	0.159	28.8	2025-09-16	
6-RA-103-08	10.97	4.57	2019-10-18	9.73	9.29	0.063	0.158	29.9	2025-09-16	
6-RA-103-09	10.97	4.57	2019-10-18	9.78	9.43	0.060	0.137	35.6	2025-09-16	
6-RA-103-10	10.97	4.57	2019-10-18	8.37	7.68	0.126	0.275	11.3	2025-09-16	
6-RA-103-11	10.97	4.57	2019-10-18	8.91	—	—	—	—	—	inaccessible – no reading
6-RA-103-12	10.97	4.57	2019-10-18	9.52	9.11	0.141	0.141	32.3	2025-09-16	
6-RA-103-13	10.97	4.57	2019-10-18	9.55	9.13	0.114	0.155	29.5	2025-09-16	
6-RA-103-14	10.97	4.57	2019-10-18	8.26	7.34	0.316	0.316	8.8	2025-09-16	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2019-10-18	6.05	5.92	0.063	0.064	37.0	2025-09-16	
3-RV-101-02	7.62	3.56	2019-10-18	5.97	5.81	0.066	0.076	29.7	2025-09-16	
3-RV-101-03	7.62	3.56	2019-10-18	5.24	4.98	0.096	0.099	14.4	2025-09-16	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-04	7.62	3.56	2019-10-18	5.58	5.39	0.083	0.074	22.1	2025-09-16	
3-RV-101-05	7.62	3.56	2019-10-18	6.11	5.91	0.066	0.068	34.4	2025-09-16	
3-RV-101-06	7.62	3.56	2019-10-18	5.55	5.39	0.083	0.086	21.2	2025-09-16	
3-RV-101-07	7.62	3.56	2019-10-18	5.07	4.80	0.117	0.112	10.6	2025-09-16	
3-RV-101-08	7.62	3.56	2019-10-18	5.86	5.81	0.075	0.076	29.7	2025-09-16	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2019-10-18	4.16	3.91	0.061	0.067	16.8	2025-09-16	
2-RV-102-02	5.54	2.79	2019-10-18	3.86	3.64	0.074	0.056	11.6	2025-09-16	
2-RV-102-03	5.54	2.79	2019-10-18	4.37	4.34	0.043	0.046	34.1	2025-09-16	
2-RV-102-04	5.54	2.79	2019-10-18	4.59	4.47	0.042	0.039	40.1	2025-09-16	
2-RV-102-05	5.54	2.79	2019-10-18	4.78	4.59	0.037	0.058	31.2	2025-09-16	
2-RV-102-06	5.54	2.79	2019-10-18	4.27	4.05	0.064	0.033	19.6	2025-09-16	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2019-10-18	7.15	6.61	0.072	0.167	16.8	2025-09-16	
4-RA-104-02	8.56	3.81	2019-10-18	7.31	7.04	0.057	0.123	26.3	2025-09-16	
4-RA-104-03	8.56	3.81	2019-10-18	7.94	7.70	0.074	0.088	44.2	2025-09-16	
4-RA-104-04	8.56	3.81	2019-10-18	7.64	7.31	0.101	0.132	26.5	2025-09-16	
4-RA-104-05	8.56	3.81	2019-10-18	7.57	7.12	0.109	0.158	21.0	2025-09-16	
4-RA-104-06	8.56	3.81	2019-10-18	7.13	6.70	0.071	0.138	20.9	2025-09-16	
4-RA-104-07	8.56	3.81	2019-10-18	6.67	6.04	0.096	0.194	11.5	2025-09-16	
4-RA-104-08	8.56	3.81	2019-10-18	7.46	7.09	0.094	0.109	30.0	2025-09-16	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2022-09-16	_____